CriptoVersus

One-page repo summary based on code, config, and solution files

What it is

CriptoVersus is a .NET 8 app that turns short-term crypto market performance into head-to-head matches where users allocate SOL-backed positions to token sides.

The repo shows a Blazor front end, an authenticated API, a background worker that builds and settles matches, and an optional Solana devnet betting program.

Who it's for

Primary persona: a crypto-native user with a Solana wallet who wants to track token-vs-token matchups, place SOL-based bets or recurring positions, and review outcomes in a wallet dashboard.

What it does

- Shows a live dashboard of top gainers, pending matches, ongoing matches, completed matches, and worker health.
- Authenticates users with Solana wallet signature login and stores JWT-backed sessions.
- Lets users invest on a token side, with support for system balance or on-chain transaction signatures.
- Tracks wallet balances, active positions, grouped history, payouts, refunds, and match-level settlement detail.
- Publishes tokenomics and protocol rules, including fee splits, auto re-entry, and on-chain program metadata.
- Provides admin/system views for privileged wallets and recent position monitoring.
- Supports match score rebuilds, score events, metric snapshots, and SignalR-driven UI refreshes.

How it works

- Web UI: `CriptoVersus` is a Blazor Server app with Razor components, session storage, JS wallet helpers, and a SignalR client.
- API: `CriptoVersus.API` exposes controllers for dashboard, matches, bets, wallet, tokenomics, login, worker status, and admin data.
- Worker: `CriptoVersus.Worker` polls Binance 24h ticker data, builds match pools, advances match states, scores outcomes, and settles capital.
- Data layer: Entity Framework Core with `EthicAIDbContext` and PostgreSQL-backed entities for users, matches, bets, currencies, ledgers, and positions.
- Realtime flow: worker/API send dashboard notifications through `/hubs/dashboard`; the web client reloads snapshots when events arrive.
- On-chain piece: `solana/criptoversus_betting` contains an Anchor devnet escrow program for match creation, betting, settlement, and payout claims.

How to run

- Prereqs from repo evidence: .NET 8 SDK, PostgreSQL connection string/user secrets for API and Worker, and optional Node/TypeScript for JS builds.
- Restore/build the projects: `dotnet restore` and `dotnet build` for `CriptoVersus.API`, `CriptoVersus.Worker`, and `CriptoVersus/CriptoVersus.Web.csproj`.
- Start the API, Worker, and Web app in separate terminals with `dotnet run --project ...` for each project.
- Open the web app, sign in with a Solana wallet, and ensure the API base URL/config points to the running API.
- Not found in repo: a local `docker-compose.yml`, seeded local database instructions, and a single documented end-to-end bootstrap script.

Evidence used: `EthicAI.sln`, project `Program.cs` files, API controllers, worker logic, Razor pages, `appsettings.json`, `CMD_SUBIR.txt`, and `solana/criptoversus_betting/README.md`.